

Assignment 8

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Electricity and Electronics EBN 111

18 May 2022

Test Information

Maximum marks:	26	Full marks:	26
Due Date:	30 May 23:59	Open/Closed book:	Open
Total number of pages (this page included)		14	

Lecturers: Prof X. Ye, Prof D. Le Roux, Mr V. Dube**Assistant Lecturers:** D. Wepener, K. Mphaga, N. Matic, J. Thiselton, L. Zheve**Important**

1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers, including complex numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your assessment, you declare the following:**

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

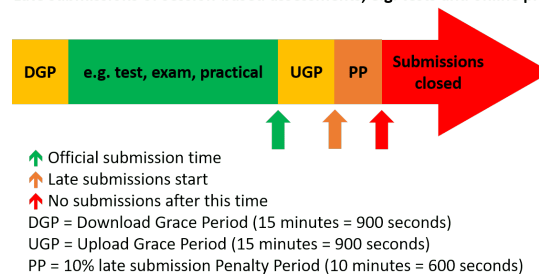
I hereby declare that the work herewith is completely my own and that no parts have been copied in any way from current or previous students or other sources.

Online assessment submission rules and time-line

The following rules and time-line apply for all tests for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the test for students to download all documents related to the test and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the test.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the test.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.ams@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers on the AMS during this period without incurring any penalties.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheet during this period, the student incurs a 10% penalty to their final test mark.
8. No late submissions are accepted after the PP.

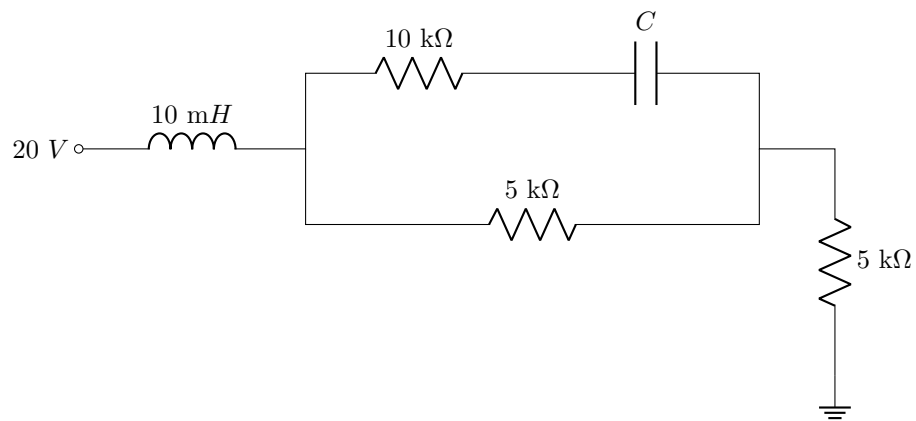
Late submissions of session-based assessments, e.g. tests and online practicals



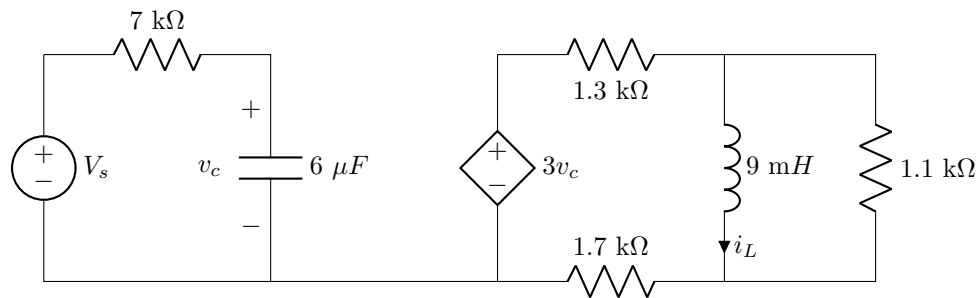
Additional guide for Complex numbers

The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

- **Correct:** $1-j2$ or $-3+j4$
 - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
 - Capital J is however incorrect, and $1-2j$ is also not acceptable.
 - If a complex number has no real part, for example $0+j2$, write $0+j2$ as the final answer.
- **Correct:**
 - With angle in degrees: $12\angle 34\deg$;
 - With angle in radians: $12\angle 0.5934$
 - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect:**
 - $1-i2$ (Don't use i , use j)
 - $1 - j2$ (Don't use spaces between characters)
 - $12 \angle 34$; $34 \deg$; (Don't use spaces between characters)
 - $j10+1$ (Don't write the imaginary part before the real part for rectangular form.)
 - $-3+4j$ (Don't write the j after the number for imaginary numbers, write $j4$ instead of $4j$)

Section 1**Question 1 [2]**Given: C .

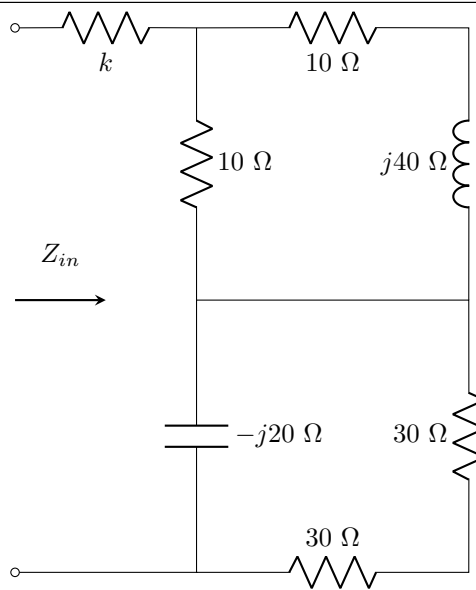
- 01** Assume steady-state conditions and determine the energy stored in the capacitor W_C [2]

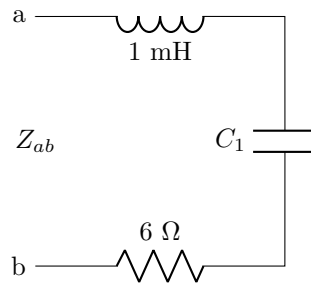
Question 2 [2]Given: V_s .**02** Assuming steady state conditions, determine the energy stored in the inductor W_L . [2]

Questions 3 to 6 [4]	
Given: X , Y , Z .	
Given the sinusoid $v(t) = X \cos(Yt + Z)$ V.	
03	Determine the angular frequency ω [1]
04	Determine the amplitude A [1]
05	Determine the phase angle θ [1]
06	Determine the frequency f [1]

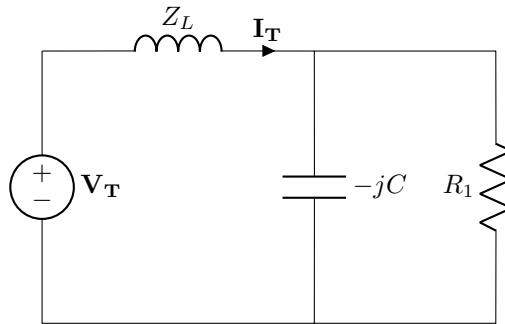
Question 7 [2]	
Given: A , B .	
Two phasors are shown below: $\mathbf{V}_1 = 8 \cos(\omega t - A^\circ)$ $\mathbf{i}_2 = 10 \sin(\omega t - B^\circ) \tag{1}$ By how many degrees is $\mathbf{i}_2$ leading $\mathbf{V}_1$? (Give your answer θ in the range from -180° to 180°)	
07	θ [2]

Question 8 [2]	
Given: X .	
Given the sinusoid: $v(t) = -4 \cos(\omega t - 240^\circ) + X \sin(\omega t + 120^\circ) V$ Write $v(t)$ in polar form as $\mathbf{V} = r\angle\theta$, with $r \geq 0$ and $-180^\circ \leq \theta \leq 180^\circ$.	
08	V [2]

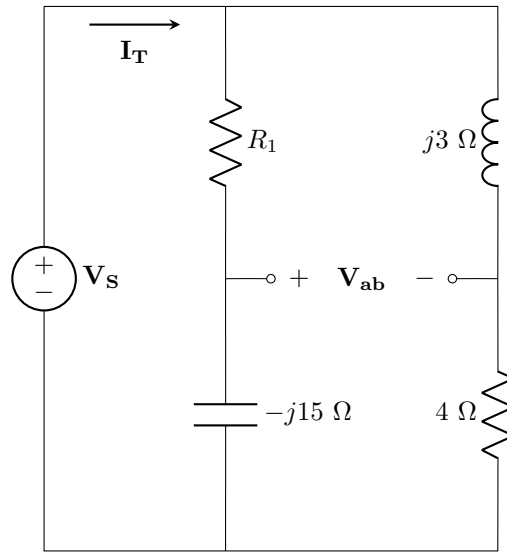
Question 9 [2]Given: k .**09** Find Z_{in} in **rectangular** form [2]

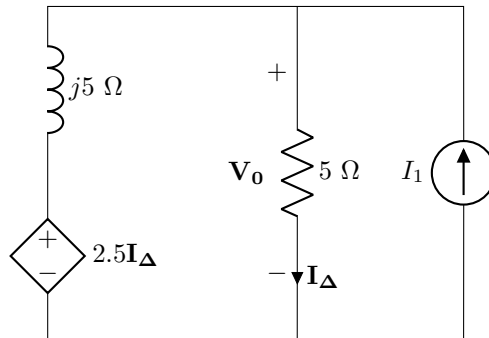
Question 10 [2]Given: C_1 .

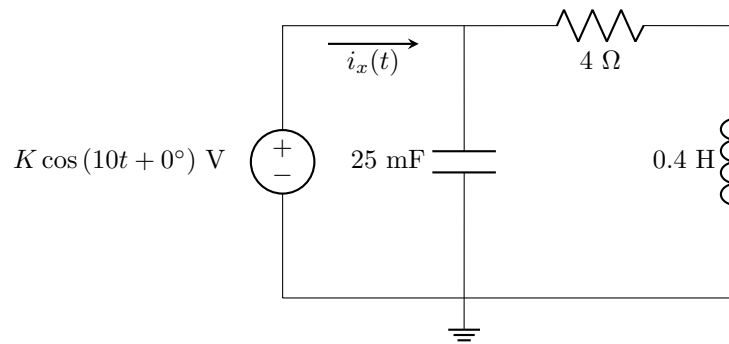
- 10** At which angular frequency ω will the circuit below have a total impedance of $Z_{ab} = 6 \Omega$ [2]

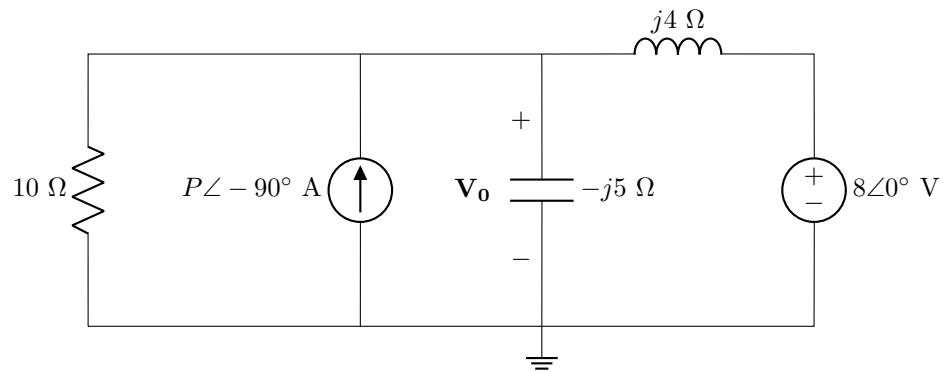
Question 11 [2]Given: C , R_1 .

- 11** Give the value of Z_L so that the phases of V_T and I_T differ by 0° . [2]

Question 12 [2]Given: R_1 , V_S .**12** Determine I_T in **rectangular** form [2]

Question 13 [2]Given: I_1 .**13** Determine V_0 in **polar** form [2]

Questions 14 to 15 [2]Given: K .Determine $i_x(t)$ in the form $i_x(t) = P \cos(\omega t + \phi)$ A.**14** Provide P where $P > 0$ [1]**15** Give ϕ where $-180^\circ \leq \phi \leq 180^\circ$ [1]

Question 16 [2]Given: P .**16** Determine V_0 in **rectangular** form [2]**Total Section 1: [26]****Grand Total : [26]**